Data Analytics Project: IGN Game Reviews Analysis

Dataset Used: ign.csv

This document outlines a structured, end-to-end data analysis project based on game reviews from the IGN dataset. The project is divided into five scenarios, increasing in difficulty from basic data handling to advanced analysis and visualization, ensuring comprehensive skill coverage.

Project Goal: To analyze game review scores, platform popularity, and genre distribution over time using the `ign.csv` dataset.

The **primary libraries/modules** required for this project are:

- **Pandas:** For all data manipulation, filtering, aggregation, and feature engineering tasks.
- **NumPy:** Specifically for numerical calculations like yearly growth (`np.diff`).
- **Visualization Library (e.g., Matplotlib):** For generating all required plots (Line Graphs, Bar Charts, Pie Charts, Stacked Bar Charts).

🎯 Project Coverage

Category	Skills Covered
Data Preprocessing	Data Loading, Missing Value Handling (Mean/Mode), Type Conversion, Column Removal
Pandas Operations	Grouping, Filtering, Aggregation, Feature Engineering (score_category, editors_choice conversion)
NumPy Calculations	Array conversion, Yearly score growth calculation (np.diff)
Visualization	Line Chart, Bar Chart, Pie Chart, Histogram, Stacked Bar Chart, Saving Plots

Scenario 1: Basic Data Loading & Cleaning (🟢 Easy)

Description:
This scenario focuses on loading and cleaning the IGN game reviews dataset using Pandas. The goal is to explore the dataset structure, identify missing values, and prepare the data for analysis. Basic preprocessing steps such as removing unnecessary columns and fixing data types are performed. This ensures the dataset is clean, consistent, and ready for further visualization or modeling.

Task	Description
1.	Load the `ign.csv` dataset using Pandas.
2.	Display the first 5 rows (`head()`), last 5 rows (`tail()`), and the shape of the dataset.
3.	Remove the unnecessary column: `"Unnamed: 0"`.
4.	Check for missing values in `score`, `genre`, and `platform`.
5.	Handle missing values: Fill the numeric column `score` with the mean, and the categorical column `genre` with the mode.
6.	Ensure correct data types: Convert `score` → float and `release_year`, `release_month`, `release_day` → integer.

```
PS C:\Users\Windows\OneDrive\Desktop\PythonTasks\Analytics\Project2> & "C:\Program Files\Python314\python.exe" c:/Users/Win
dows/OneDrive/Desktop/PythonTasks/Analytics/Project2/app.py
   Unnamed: 0 score_phrase                                        title  ... release_year release_month  release_day
0           0      Amazing                             LittleBigPlanet PS Vita  ...         2012             9           12
1           1      Amazing  LittleBigPlanet PS Vita -- Marvel Super Hero E...  ...         2012             9           12
2           2        Great                                Splice: Tree of Life  ...         2012             9           12
3           3        Great                                             NHL 13  ...         2012             9           11
4           4        Great                                             NHL 13  ...         2012             9           11

[5 rows x 11 columns]
       Unnamed: 0 score_phrase                                    title  ... release_year release_month  release_day
18620       18620         Good                   Tokyo Mirage Sessions #FE  ...         2016             6           29
18621       18621      Amazing           LEGO Star Wars: The Force Awakens  ...         2016             6           29
18622       18622     Mediocre  Star Ocean: Integrity and Faithlessness  ...         2016             6           28
18623       18623  Masterpiece                                     Inside  ...         2016             6           28
18624       18624  Masterpiece                                     Inside  ...         2016             6           28

[5 rows x 11 columns]
(18625, 11)
[8.5        6.16174497 6.56634146 6.93067847 6.98348387 6.69358852
 7.12049419 6.97852564 7.19824561 7.17071214 7.22242647 6.73286424
 6.63254658 6.44814621 6.84036752 6.93286867 7.05032619 7.41832669
 7.46048851 7.46759777 7.65945205 7.57345133]
```

Figure 1.1 Displaying outputs from head(), tail(), shape functions

Conclusion:

In this scenario, the IGN dataset was successfully loaded and cleaned using Pandas. Missing values were identified and handled using statistical imputation methods such as mean and mode. An unnecessary index column was removed to improve data structure. Additionally, data types were corrected to ensure consistency across numerical and categorical fields. This preprocessing step ensures the dataset is ready for further analysis and visualization.

Scenario 2: Line Graph (Score Trend) + Save (🟢 Easy)

Description:
This scenario analyzes how video game review scores change over time using the cleaned IGN dataset. The data is grouped by release year to calculate the average score for each year. The results are converted into NumPy arrays for visualization. A line graph is created to observe trends in game quality over the years, and the final plot is saved as an image file for reporting purposes.

Task	Description
1.	Group data by `release_year`.
2.	Calculate the average score per year using Pandas.
3.	Convert the resulting average scores and years into NumPy arrays.
4.	Plot a line graph with `release_year` on the X-axis and average score on the Y-axis.
5.	Add the title: "Average Game Score Over Years" and proper axis labels.
6.	Save the graph: `plt.savefig("avg_score_trend.png")`.

Graph:
The graph displays the trend of average video game review scores over different release years. The X-axis represents the release year, while the Y-axis represents the average score. The line graph helps visualize whether game quality has improved, declined, or remained stable over time.

Figure 2.1: Line Chart of Every Year's Average Score

Conclusion:
In this scenario, the dataset was analyzed to understand how game scores vary across different years. The data was grouped by release year, and the average score for each year was calculated. These values were then converted into NumPy arrays for visualization. A line graph was plotted to show trends in game quality over time. Finally, the graph was saved as an image file for documentation. This analysis helps in identifying long-term patterns in game reviews.

Scenario 3: Filtering + Bar Chart + Save (🟡 Medium)

Description:
In this scenario, the dataset is filtered to focus only on high-rated games (score > 7). The goal is to identify which gaming platforms host the highest number of well-reviewed games. After filtering, the data is grouped by platform and aggregated to count the number of high-rated games per platform. The top 10 platforms are selected and visualized using a bar chart for easy comparison.

Task	Description
1.	Filter the dataset to include only games where `score > 7`.
2.	Count the number of high-rated games per platform.
3.	Select the top 10 platforms using Pandas.
4.	Convert the platform counts into NumPy arrays.
5.	Plot a bar chart with `platform` on the X-axis and count of games on the Y-axis.
6.	Rotate x-axis labels for readability.
7.	Save the graph: `plt.savefig("top_platforms_bar.png")`.

Graph:
The bar chart displays the top 10 platforms with the highest number of games rated above 7. The X-axis represents different platforms, while the Y-axis shows the count of high-rated games. Rotated labels improve readability and allow easy comparison across platforms.

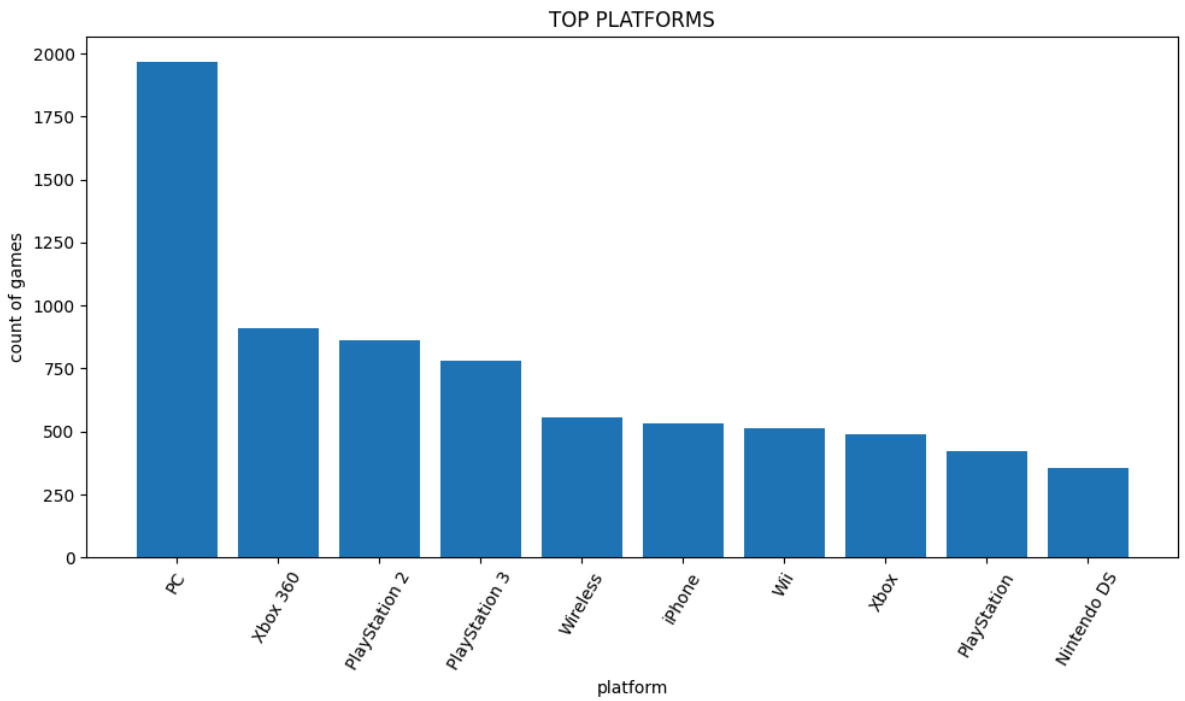

Figure 3.1: Bar Chart of Top 10 Platforms

Conclusion:

This analysis highlights the platforms that consistently host high-quality games. It shows that certain platforms dominate in terms of highly rated titles, indicating strong developer support and user engagement. The visualization makes it easy to compare platform performance and identify industry leaders.

Scenario 4: Aggregation + Pie Chart + Save (🟡 Medium)

Description:
This scenario focuses on analyzing the distribution of games across different genres. The dataset is grouped by genre to count the number of games in each category. The top 5 most frequent genres are selected for visualization. A pie chart is used to represent the proportional distribution of these genres.

Task	Description
1.	Count the total number of games per `genre`.
2.	Select the top 5 genres using Pandas.
3.	Prepare labels (genres) and values (counts) for the pie chart.
4.	Plot a pie chart using `genre` as labels and `count` as values.
5.	Add percentage labels using `autopct` (e.g., '%.1f%%').
6.	Save the graph: `plt.savefig("genre_distribution.png")`.

Graph:
The pie chart illustrates the percentage share of the top 5 game genres. Each slice represents a genre, and the size of the slice corresponds to the number of games in that category. Percentage labels (autopct) provide a clear understanding of distribution.

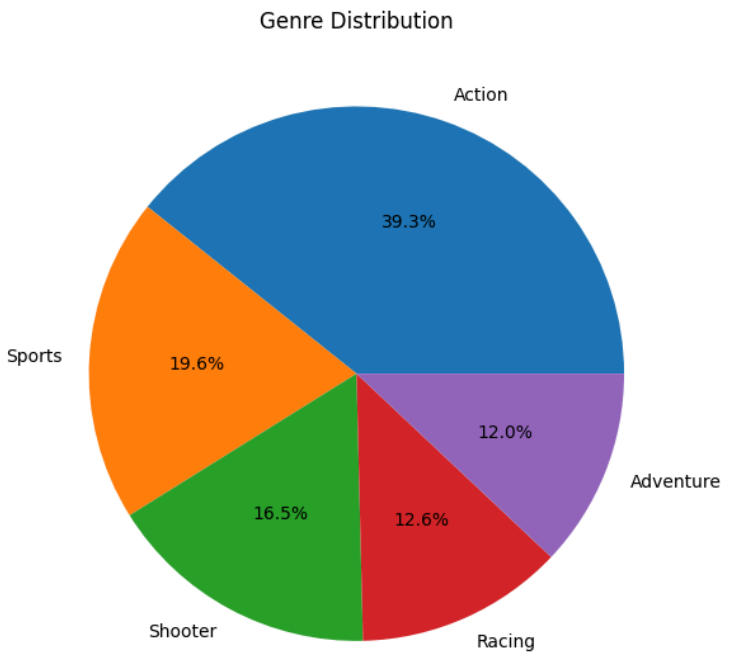

Figure 4.1: Pie Chart of Genre Distribution

Conclusion:
The analysis reveals which genres are most popular in the dataset. A few genres dominate the majority of games, suggesting trends in player preferences and industry focus. This insight can help developers and analysts understand market demand.

Scenario 5: Advanced Analysis + Multiple Graphs (🔴 Hard)

Description:
This scenario performs advanced data analysis using feature engineering, NumPy operations, and multiple visualizations. A new column (`score_category`) is created to classify games based on their scores. The `editors_choice` column is converted into numeric format for analysis. Yearly score growth is calculated using NumPy's `np.diff()`. Multiple visualizations—including a line graph, stacked bar chart, and histogram—are used to uncover deeper insights into review patterns.

Task	Description
1.	Create a new column `score_category`: "Excellent" for score ≥ 9, "Good" for $7 \le$ score < 9, and "Average" for score < 7.
2.	Convert the `editors_choice` column: 'Y' $\rightarrow 1$, 'N' $\rightarrow 0$.
3.	Use NumPy's `np.diff()` to calculate the **yearly score growth** based on the average yearly scores from Scenario 2.
4.	Plot a **Line Graph** showing the trend of average score per `release_year`.
5.	Plot a **Stacked Bar Chart** showing the count of `score_category` per `release_year`.
6.	Plot a **Histogram** showing the distribution of `score`.
7.	Save the graphs: `plt.savefig("score_trend.png")`, `plt.savefig("score_category_stacked.png")`, `plt.savefig("score_distribution.png")`.
8.	Identify: Which years had the highest scores, whether high scores increased over time, and if `editors_choice` correlates with high scores.

Graph:

1. Line Graph (Score Growth):

Shows how average game scores change year by year. It highlights fluctuations and trends in review scores over time.

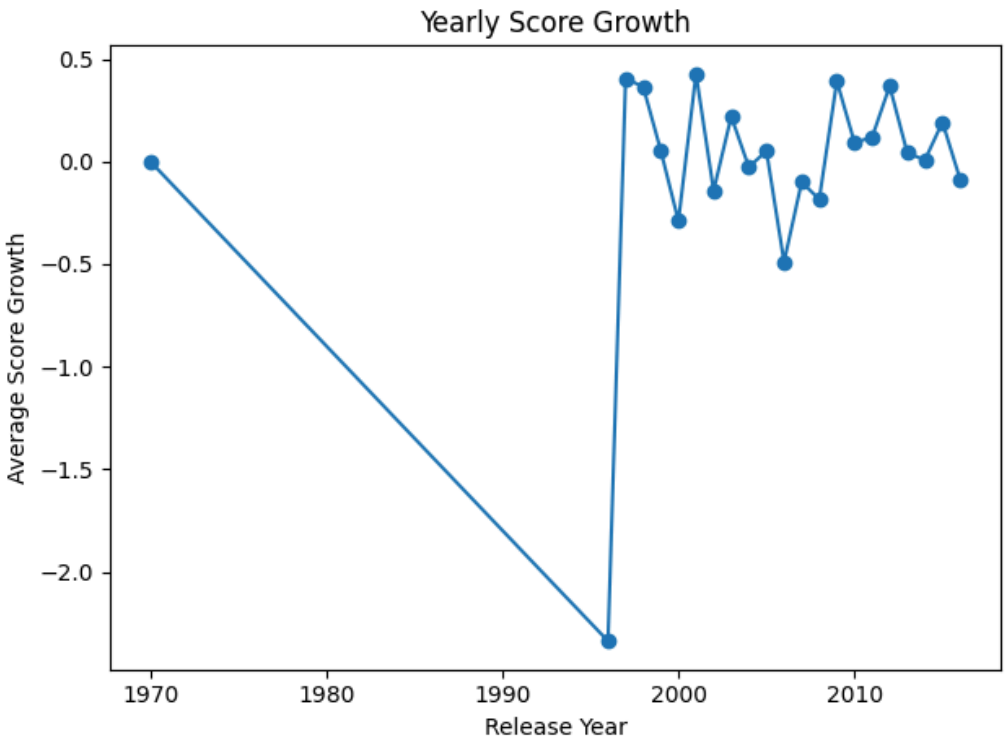

Figure 5.1:Line Graph of Yearly Score Growth

2. Stacked Bar Chart:

Displays the count of "Excellent", "Good", and "Average" games per year. It helps compare how score categories evolve over time.

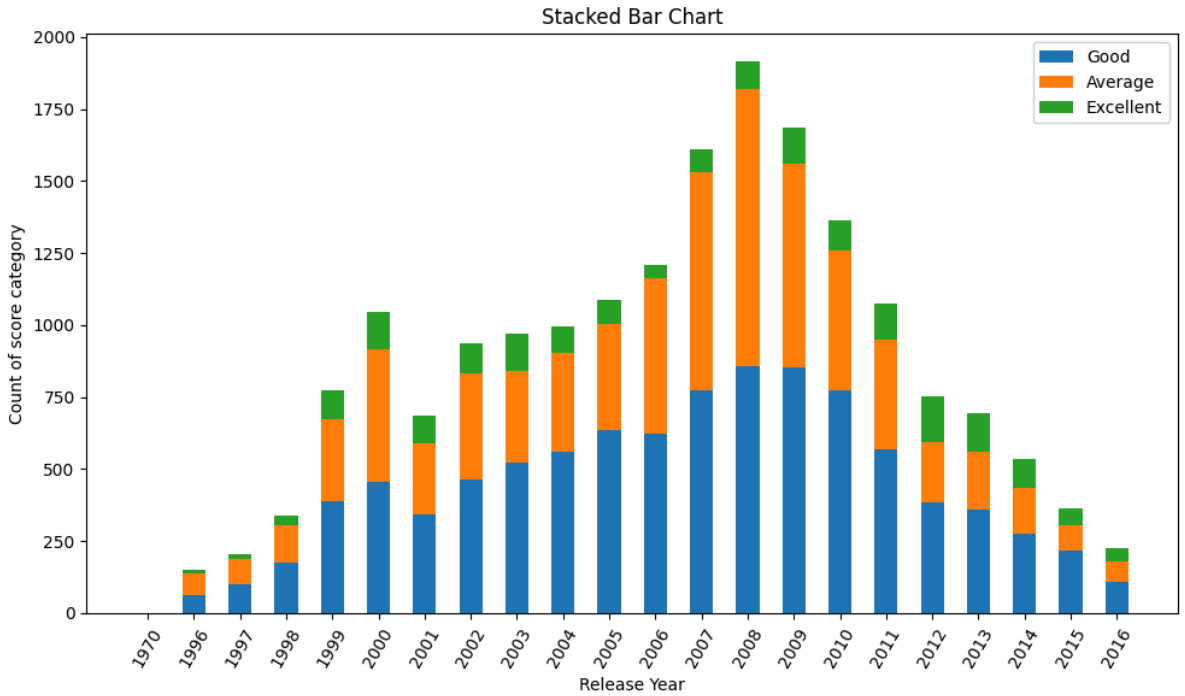

Figure 5.2: Stacked Bar Chart of Score Category per Year

3. Histogram:

Shows the overall distribution of game scores, indicating how frequently different score ranges occur.

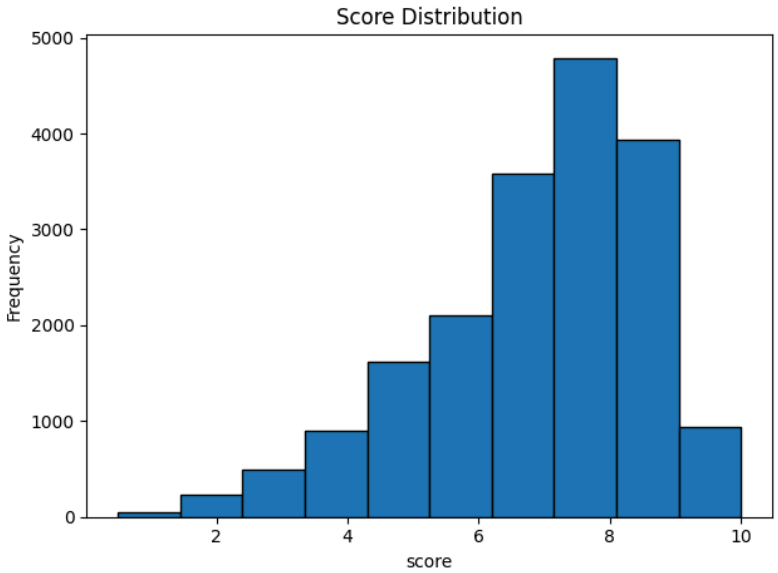

Figure 5.3: Histogram of Score Distribution

Conclusion:
This scenario provides deeper insights into game review patterns. Most games fall into the "Good" category, while "Excellent" games are relatively rare. The stacked bar chart shows an increase in high-rated games over time, especially after 2000. The histogram confirms that scores are concentrated between 7 and 9. Additionally, the analysis shows a strong relationship between high scores and editor's choice selections.

Final Conclusion:

This project successfully demonstrates a complete data analysis workflow using the IGN game reviews dataset. Starting from data cleaning and preprocessing, the analysis progressed through trend visualization, platform comparison, and genre distribution. Advanced techniques such as feature engineering and NumPy-based calculations were applied to extract deeper insights. The results show that game quality and volume improved significantly after the early 2000s, with certain platforms and genres dominating the industry. Overall, the project highlights how data analysis can uncover meaningful trends in gaming data.

What you have learned

- How to clean and preprocess real-world datasets using Pandas
- Handling missing values using mean and mode
- Performing grouping, filtering, and aggregation operations
- Converting data into NumPy arrays for numerical analysis
- Using `np.diff()` to calculate trends and growth
- Creating different types of visualizations:
    - Line charts
    - Bar charts
    - Pie charts
    - Histograms
    - Stacked bar charts
- Saving plots for reporting and presentation
- Interpreting data and extracting meaningful insights